

Classical Information Theory

Shannon Entropy, Classical Typicality, and Coding Theorems

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Introduction: An example

The Task

Let's say Alice is trying to communicate a message to Bob, from a source that randomly chooses from four symbols $\mathcal{X} = \{a, b, c, d\}$ according to the following distribution:

$$\begin{aligned} P(a) &= \frac{1}{2} & P(b) &= \frac{1}{4} \\ P(c) &= \frac{1}{8} & P(d) &= \frac{1}{8} \end{aligned}$$

Let's say the “channel” Alice must use is noiseless, and it only takes binary bits as inputs. One might ask two main questions:

- How should Alice encode these 4 messages ?
- What's the most efficient way to do it ?
- How do you quantify this “efficiency” ?

Naive Encoding

The obvious way to do it, would be to use 2 bits per message, since there are a total of 4 messages. So, Alice could use the following *code*:

$$\begin{aligned}a &\rightarrow 00, & b &\rightarrow 01, \\c &\rightarrow 10, & d &\rightarrow 11\end{aligned}$$

The average number of bits being used per message is given by:

$$\# \text{ of bits} = \sum_{x \in \{a,b,c,d\}} P(x) \cdot 2 = 2$$

However, notice that we did not take advantage of the fact that some of the symbols are way more likely than the others. We could assign shorter (longer) strings for more (less) likely messages.

Compressed Encoding

Consider the following coding scheme:

$$\begin{aligned} a &\rightarrow 0, & b &\rightarrow 10, \\ c &\rightarrow 110, & d &\rightarrow 111 \end{aligned}$$

This code is still decodable at Bob's end, and is from a special class of codes known as a *Prefix Code*.

The average number of bits used per message is then given by:

$$\begin{aligned}
\# \text{ of bits} &= \sum_{x \in \{a,b,c,d\}} P(x) \cdot n(x) \\
&= P(a)n(a) + P(b)n(b) + P(c)n(c) + P(d)n(d) \\
&= \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4} = 1.75 < 2 \\
&= \frac{1}{2} \log_2(2) + \frac{1}{4} \log_2(4) + \frac{1}{8} \log_2(8) + \frac{1}{8} \log_2(8) \\
&= \sum_{x \in \{a,b,c,d\}} P(x) \log_2 \left(\frac{1}{P(x)} \right) \\
&= - \sum_{x \in \{a,b,c,d\}} P(x) \log_2(P(x))
\end{aligned}$$

Hence, the average number of bits required in this coding scheme is 1.75 which is strictly better than our previous coding scheme.

Can we do better?

The answer to this question is infact NO, as proved by Shannon in his 1948 paper, which particularly via his source coding theorem. Which we will come to eventually.

But first, we must look at some definitions.

Classical Information and Entropy

Shannon Entropy: Definition

Consider a random variable X with the associated probability distribution $P_X(x)$ where each realisation of X is from some alphabet $\mathcal{X}$. Then we have the following definitions.

Definition 2.1 (Information content): The *information content* of a particular realisation $x \in \mathcal{X}$ is defined as

$$i(x) = -\log_2(P_X(x))$$

As you can see, lower the probability of some realisation, higher the information content. It's intuitively helpful to think about this as a measure of “surprise”.

Definition 2.2 (Entropy): The *Entropy* of a random variable X is defined as the expected information content of the random variable X , which is given by

$$H(X) = - \sum_{x \in \mathcal{X}} P_X(x) \log_2(P_X(x))$$

Again, it's helpful to think of this as the expected amount of “surprise” from the random variable X .

Now we will look at some properties of this quantity.

Shannon Entropy: Properties

Property 2.1 (Non-negativity): $H(X) \geq 0$ for any discrete random variable X .

Property 2.2 (Maximum Value): For any discrete random variable X , we have

$$H(X) \leq \log_2 |\mathcal{X}|$$

And equality holds iff X is a uniform random variable.

Conditional Entropy

Let's say Alice has access to some random variable X and Bob has access to some random variable Y . In general X and Y might not be independent. So, one could say that Bob possesses some “side-information” about X via Y . To quantify this, we define the following quantities.

Definition 2.3 (Conditional Information Content): The conditional information content of a realisation of X given some realisation y of Y , is defined as

$$i(x|y) = -\log_2(P_{X|Y}(x|y))$$

Conditional Entropy

Similarly, we can also define conditional entropy of X given Y

Definition 2.4 (Conditional Entropy): Let X and Y be two discrete random variables. The conditional entropy $H(X|Y)$ is defined as the expected conditional information content, where expectation is with respect to both X and Y .

$$\begin{aligned} H(X|Y) &= - \sum_{x \in X, y \in Y} P_{X,Y}(x, y) \log_2(P_{X|Y}(x|y)) \\ &= - \sum_{y \in Y} P_Y(y) \sum_{x \in X} P_{X|Y}(x|y) \log_2(P_{X|Y}(x|y)) = \sum_{y \in Y} P_Y(y) H(X | Y = y) \end{aligned}$$

More Properties

Property 2.3 (Conditioning does not increase entropy): For any two discrete random variables X and Y ,

$$H(X) \geq H(X|Y)$$

Equality only occurs when X and Y are independent.

This property is quite intuitive, since your surprise from X when only know X will of course be (at least) higher than if you knew something about Y , which might share some correlations with X .

Joint Entropy

What if Bob knew neither X or Y ? Then one would have to look the joint entropy of the random variables, which is simply the entropy of the joint random variable (X, Y) .

Definition 2.5 (Joint Entropy): Consider two discrete random variables X and Y with the joint probability distribution $P_{X,Y}(x, y)$. The *Joint Entropy* of X and Y is given by

$$H(XY) = - \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} P_{X,Y}(x, y) \log_2(P_{X,Y}(x, y))$$

Even more properties

Property 2.4 (Subadditivity): If you have n random variables $\{X_i\}_{i=1}^n$, then we have

$$H(X_1 \dots X_n) \leq H(X_1) + \dots + H(X_n)$$

Property 2.5 (Chain Rule): Again for n random variables $\{X_i\}_{i=1}^n$, then the following relation holds

$$H(X_1 \dots X_n) = H(X_1) + H(X_2|X_1) + \dots H(X_n|X_{n-1}, \dots, X_1)$$

Mutual Information

We now introduce an information measure that quantifies the common or “mutual” information between two parties i.e. random variables.

Definition 2.6 (Mutual Information): Let X and Y be discrete random variables with the joint probability distribution $P_{X,Y}(x,y)$. Then the *mutual information* $I(X;Y)$ is defined as,

$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(XY) \end{aligned}$$

Mutual Information

$I(X; Y)$ quantifies by how much the uncertainty about one random variable decreases when one knows the other. If Bob knows Y , then his uncertainty about X is now given by $H(X|Y)$, which is $I(X; Y)$ less than his uncertainty about X is he didn't know Y , which would be given by $H(X)$.

We can express the mutual information in terms of the joint and marginal probability distributions as follows

$$I(X; Y) = \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} P_{X,Y}(x, y) \log_2 \left(\frac{P_{X,Y}(x, y)}{P_X(x) \cdot P_Y(y)} \right)$$

Mutual Information

Property 2.6 (Mutual Information is non-negative): For any two random variables X and Y , we have

$$I(X; Y) \geq 0$$

And it's zero iff X and Y are independent.

Classical Typicality

Definition(s)

Shannon used the idea of (weak) typicality to prove his two coding theorems. In this section we will introduce the idea of typicality and look at some relevant properties.

Consider a set of iid random variables $X_1, \dots, X_n$. We will denote a realisation of these iid distribution by x^n

Definition 3.1 (Typical Sequence): A sequence x^n is called δ typical if the following is satisfied:

$$\left| -\frac{1}{n} \log_2(P_{X^n}(x^n)) - H(X) \right| \leq \delta$$

Definition(s)

Definition 3.2 (Typical Set): The δ – typical set $T_\delta^{X^n}$ is defined as the set of all δ – typical sequences.

Properties

Property 3.1 (Any sequence is almost certainly typical): Invoking the weak law of large numbers we can clearly see that, $\forall \varepsilon \in (0, 1)$, $\delta > 0$ we will have

$$P(X^n \in T_\delta^{X^n}) \geq 1 - \varepsilon$$

for sufficiently large $n \in \mathbb{N}$.

Property 3.2 (Equipartition): Just by manipulating the definition of $T_\delta^{X^n}$ one can show that

$$2^{-n(H(X)+\delta)} \leq P_{X^n}(x^n) \leq 2^{-n(H(X)-\delta)}$$

All typical sequences occur with approximately the same property for large enough n .

Properties

Property 3.3 (Cardinality): Using the previous two properties one can show that the cardinality of $T_\delta^{X^n}$ satisfies the following bounds

$$(1 - \varepsilon)2^{n(H(X)-\delta)} \leq |T_\delta^{X^n}| \leq 2^{n(H(X)+\delta)}$$

We now have the required mathematical machinery to prove the first of the two coding theorems.

Joint Typicality

- Definitions
- Properties

Shannon's Source-Coding Theorem

Setting up the problem: Compression

Shannon's Source coding theorem is in regards to the rate of compression. Let's say we have a set of n iid random variables. An encoder is encoding this information into binary strings of length l_n . Formally, the encoding is the following map:

$$E : \mathcal{X}^n \rightarrow \{0, 1\}^{l_n}$$

The decoder then does the reverse,

$$D : \{0, 1\}^{l_n} \rightarrow \mathcal{X}^n$$

We define the probability of error $P_e^{(n)}$ as

$$P_e^{(n)} = \Pr\{D(E(X^n)) \neq X^n\}$$

The compression scheme is reliable if

$$\lim_{n \rightarrow \infty} P_e^{(n)} = 0$$

and the asymptotic rate of compression is defined as

$$R = \lim_{n \rightarrow \infty} \frac{l_n}{n}$$

Our goal is to find the least possible R .

If we go back to our example, we can easily compute that the rate we achieved for the second encoding is essentially $H(X)$ where X is the random variable with the associated alphabet $\mathcal{X} = \{a, b, c, d\}$ and associated probability distribution P_X defined accordingly. Shannon's source-coding theorem shows that this is the best achievable rate of reliable compression.

Theorem Statement

Theorem 4.1 (Source-coding Theorem): The entropy of a random variable X is the smallest achievable rate for compression.

$$H(X) = \inf\{R : R \text{ is an achievable rate of reliable compression}\}$$

We prove this in two parts. First we show that any rate R that is greater than $H(X)$ is an achievable rate. Then, we show that any rate R that is less than $H(X)$ leads to a coding scheme in which the probability of correct decoding goes to 0 as $n \rightarrow \infty$.

Theorem Statement

Proof: Consider some $R > H(X)$. Then, $\exists \delta > 0$ such that, $H(X) + \delta < R$. Then, we have

$$|T_\delta^{X^n}| \leq 2^{n(H(X)+\delta)} \leq 2^{nR}$$

So, using nR bits we can index every δ – typical sequence i.e. $\exists f : T_\delta^{X^n} \rightarrow \{0, 1\}^{nR}$. Now, consider the following compression scheme: If $x^n \in T_\delta^{X^n}$, then we encode the index using nR bits and if $x^n \notin T_\delta^{X^n}$, we encode it as 0^n . The decoding process is the inversion of this map into the set of typical sequences. We ignore all atypical sequences when decoding.

Hence the probability of error is bounded as,

$$0 \leq P_e^{(n)} \leq \Pr[X^n \notin T_\delta^{X^n}]$$

But by the weak law of large numbers we have,

$$\begin{aligned}\lim_{n \rightarrow \infty} \Pr[X^n \notin T_\delta^{X^n}] &= 0 \\ \Rightarrow \lim_{n \rightarrow \infty} P_e^{(n)} &= 0\end{aligned}$$

Therefore, any $R > H(X)$ is an achievable rate of reliable compression.

Now, consider some $R < H(X)$. Again, $\exists \delta > 0$ such that $R + \delta < H(X)$. Observe that if we are using nR bits, there are at most 2^{nR} different sequences that the decoder can successfully distinguish. We know that $\forall x^n \in T_\delta^{X^n}$,

$$P_{X^n}(x^n) \leq 2^{-n(H(X)-\delta)}$$

Let's assume that all the 2^{nR} codewords are perfectly decodable and are all the highest probable typical sequences. Furthermore, let's assume we can also decode all atypical sequences perfectly. This gives us the following bound for the probability of successful decoding

$$\begin{aligned} 0 \leq 1 - P_e^{(n)} = P_c^{(n)} &\leq 2^{nR} 2^{-n(H(X) - \delta)} + \Pr[X^n \notin T_\delta^{X^n}] \\ &= 2^{n(R - H(X) + \delta)} + \Pr[X^n \notin T_\delta^{X^n}] \end{aligned}$$

Observe that $R - H(X) + \delta < 0$ by definition, and hence that term vanishes as $n \rightarrow \infty$. The second also vanishes as $n \rightarrow \infty$ by the weak law of large numbers. Hence, we finally have,

$$\lim_{n \rightarrow \infty} P_c^{(n)} = 0 \Rightarrow \lim_{n \rightarrow \infty} P_e^{(n)} = 1$$

Hence, reliable compression at a rate $R < H(X)$ is not achievable. This concludes our proof. ■

Noisy Channel Coding Theorem

Classical Channels

We will be dealing with discrete memory-less channels here, also called DMCs in short.

Definition 5.1 (Classical Channel): A classical channel with input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$ is defined as the map $\mathcal{N} : \mathcal{X} \rightarrow \mathcal{Y}$, characterised by the conditional “jump” probabilities $P_{Y|X}(y|x)$.

Given some probability distribution over the inputs (say some random variable X), the Channel is essentially “creating” a random variable Y , with a given conditional probability distribution.

Memoryless-ness of a channel means that each successive use of the channel is not dependant on the previous use, i.e. the jump probabilities stay the same.

Capacity of a Classical Channel

TODO

Theorem Statement

Theorem 5.1 (Channel Capacity Theorem): The capacity of a discrete memoryless channel is given by the maximum mutual information of the channel.

$$C(\mathcal{N}) = \max_{P_X} I(X; Y)$$

The optimization is over all possible input distributions.